



Sex- and experience-dependent regulation of synaptic protein turnover

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Synaptic transmission can be tuned through plasticity mechanisms that regulate synaptic strength, structure, and number. *In vivo* measurements demonstrate remarkable spine dynamics, with subsets of synapses persisting for months. This correlates with the longevity of certain memories, which can persist for an organism's lifetime. The molecular basis supporting the long-term stability of specific synapses and the long-term durability of memories remains unknown. At the protein level, most proteins persist for a relatively short amount of time before they are degraded and replaced with new molecules. However, recent work has identified a population of proteins, including those present at the synapse, that are exceptionally long-lived. It has been speculated that long-lived proteins (LLPs) could contribute to long-term synapse stability, function, and memory. Here, we used stable isotope labeling in mammals to first identify LLPs in the post synaptic density (PSD) of the hippocampus and subsequently determine if protein turnover rates varied by sex or following learning. We identified synaptic LLPs and found that both sex and experience can regulate synaptic protein turnover rates. We identified sex-dependent changes in protein turnover rates in autism spectrum disorder risk genes, including increased stability of Gabrg2, a GABA-A receptor subunit, in male mice. Furthermore, we observed stabilization of a subset of PSD proteins, such as Shank3, following contextual fear conditioning. We propose that sex- and experience-dependent changes in protein turnover rates could help explain sex-differences in psychiatric risk and aid our understanding of the molecular mechanisms that support learning and memory.

protein stability | synapse | proteomics | sex differences

Individual proteins have finely tuned rates of synthesis and degradation, with lifetimes spanning several orders of magnitude. Post mitotic cells, including neurons, have been shown to contain a subset of particularly stable proteins, often referred to as long-lived proteins (LLPs). In brain tissue, examples of LLPs include nuclear pore complex proteins (1), components of the extracellular matrix, as well as documented synaptic proteins (2–5). While the contribution of LLPs to synapse function remains largely enigmatic, notable examples have been linked to synapse function and animal behavior. For example, the collapsing response mediator protein (CRMP) family members were previously identified as being particularly stable at the synapse (4). Crmp2 KO mice exhibit impaired synaptic plasticity, alterations in dendrite morphology, and changes in behaviors relevant to psychiatric disorders (5), and mice lacking Crmp5 show abnormalities in dendritic development, cell body size, as well as long term depression synaptic plasticity deficits in Purkinje cells (6), highlighting that loss of these stable proteins has dramatic effects on neuronal function.

Neuronal cells have some unique challenges to overcome as they typically exhibit long-range axonal projections and extensive dendritic arborizations, for which they need to maintain integrity and proper function for the lifetime of the organism. With the development of *in vivo* imaging methodologies, it has been shown that individual synapses have huge variability in their longevity, with some very transient and others persisting for months (7). The stability of some synapses is quite remarkable and readily exceeds the estimated turnover rate of most synaptic proteins. With synaptic plasticity thought to be a correlate of learning and memory a paradox emerges: how can changes in synapse function (i.e., LTP and spine dynamics) be maintained longer than the lifetimes of the proteins that form the constituent parts of the synapse? The synaptic tagging hypothesis posits that activated synapses contain a biomolecular tag that promotes subsequent—and long term—sequestration of proteins which support LTP (8). LLPs have been identified at the synapse (4), raising the question as to whether a subset of exceptionally stable proteins could represent a “synaptic tag” to support the long-term stability of specific synapses, or even the maintenance of long-term memories.

Significance

While most proteins are rapidly synthesized and degraded, a subset of synaptic proteins exhibit remarkable stability. The role of these long-lived synaptic proteins is still enigmatic but could regulate synapse function. Furthermore, it is not well understood if the stability of individual proteins can be dynamically regulated by experience, or whether there are basal sex differences in protein turnover rates. We address this using high-resolution mass spectrometry to calculate the relative turnover rates of synaptic proteins in the hippocampus. We confirm the existence of a subset of exceptionally long-lived proteins and find others that exhibit both sex and experience dependence changes in their turnover rates. These findings have implications for the understanding of synapse function and psychiatric risk.

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Protein turnover rates are frequently assessed using stable isotope labeling in mammals (SILAM) coupled with mass spectrometry, wherein protein degradation and subsequent synthesis lead to changes in the relative abundance of heavy isotopes within individual proteins. Previous research shows that protein turnover rates are dependent on cellular environment as the presence of Glia has been shown to increase the turnover rates of neuronal proteins *in vitro* (9). Additionally, the experience of environmental enrichment has been shown to alter protein turnover rates of synaptic proteins *in vivo* (3, 4). Currently, whether protein turnover rates are altered by sex remains unknown. This is a pressing issue because there are well documented sex differences in the prevalence of psychiatric disorders. For example, autism spectrum disorder (ASD) is more readily diagnosed in males (10). While acknowledging that this could also be attributed to missed diagnosis in women, it also suggests possible biological sex differences underlying ASD. SFARI has collated ASD risk genes (<https://gene.sfari.org/>), many of which are established synaptic proteins.

In the present study, we sought to document how turnover rates of the hippocampal synaptic proteome changes with sex, or after a fear-learning behavioral task. Our analysis focused on the post-synaptic density (PSD) of the hippocampus. As expected, we identified a subset of proteins with long half-lives. Curiously, we found sex differences in synaptic protein turnover includes many ASD risk genes. Additionally, we show that a subset of synaptic proteins tune their turnover rates following contextual fear conditioning (cFC).

Results

Identification of Stable Synaptic Proteins. We conducted a SILAM experiment to characterize the relative turnover rates of proteins enriched at the PSD. To promote maximal isotope labeling, chow containing Lys-¹³C₆ was given to a breeding pair so labeling could begin *in utero*. When the resulting offspring reached 10 wk, they were switched to a control diet with regular Lys-¹²C₆ to initiate a “chase” period (Fig. 1A). Tissue was collected at different time points and biochemical fractionation performed on hippocampal tissue to extract the PSD fraction (Fig. 1B). Proteins extracted from PSD were analyzed by mass spectrometry, resulting in the identification of a total of 4,145 proteins. At time point 0 the average relative isotope abundance (RIA) for each protein was 0.87 for the PSD fraction, indicating very high levels of labeling, with 2,961 proteins having a starting RIA > 0.8 (Fig. 1C).

Previously, Fornasiero et al. measured the stability of proteins in the brain by implementing a double exponential function that considers the recycled pool of amino acids containing the heavy isotope, providing a precise calculation of protein lifetime (3). We applied this method to our data, using their reported constant(s). Unexpectedly, we observed that many proteins in our dataset had estimated half-lives with values close to zero (SI Appendix, Fig. S1A). Of note, the constant(s) used by Fornasiero et al., were derived from male mice only. To see if sex impacted our implementation of their formula, we split our data into male and female animals. There was a larger proportion of close-to-zero protein half-lives within the isolated female dataset (SI Appendix, Fig. S1B). Because the constant used in this calculation method is acutely sensitive to differences in metabolism rates likely different in male and female mice, a separate calibration experiment would be required to calculate appropriate constants for female animals. Indeed, it is likely that any experimental group where metabolism could be different will require its own calibration experiments. Therefore, to circumvent this confound,

we proceeded by evaluating turnover rates using a simple exponential decay function. At each time point during the chase period the RIA was calculated for each protein. By fitting an exponential decay function to the RIA values, we estimated the half-life of each protein, where slower RIA decay indicates proteins undergo slower degradation (Fig. 1D). This revealed proteins with vastly different turnover times, exemplified by Vcan and Psd95 (Fig. 1E). Reassuringly, we observed a strong positive and highly significant correlation between protein half-lives in our PSD dataset (estimated by simple exponential decay) and half-lives reported by Fornasiero et al., in the synaptosome fraction of both cortex and cerebellum tissue (SI Appendix, Fig. S1C and D). We acknowledge that we are likely overestimating protein half-lives and therefore focus on the *relative* turnover rates between proteins.

We identified a subset of PSD proteins with exceptionally slow relative turnover rates, including proteins found in our previous study such as Vcan, Traf3, and Rras2 (4), as well as previously unidentified proteins (Fig. 1F and SI Appendix, Fig. S1E and Table S1). We conducted a STRING analysis of the top 50 most stable proteins. These proteins were significantly connected and were enriched for GO-terms such as postsynaptic intermediate filament cytoskeleton, perineuronal net, and collagen-containing extracellular matrix (Fig. 1G). We conducted a manual evaluation of relative half-lives of different ionotropic glutamate receptors (Fig. 1H), metabotropic glutamate receptors (Fig. 1I), membrane-associated guanylate kinase family members (Fig. 1J), SH3 and multiple ankyrin repeat domain containing proteins (Fig. 1K), and GABA receptors (Fig. 1L). It was interesting to observe differences in the half-lives of proteins in the same family, such as the relative stability of GluA4 (Fig. 1H), an AMPA receptor subunit particularly enriched in GABAergic inhibitory interneurons (11), as well as the reduced stability of Sap102 (Fig. 1J), a protein known to be more mobile at synapses compared to other MAGUKs (12).

Sex Differences in Protein Turnover. To our knowledge, previous studies looking at protein turnover rates have not considered sex differences. We were interested to probe for basal changes in protein turnover based on sex, as there are well documented sex differences in the prevalence of psychiatric disease (13) and neurological disorders (14). Therefore, we split our dataset into male and female animals and recalculated relative protein turnover rates. When analyzed separately there was no difference in the initial RIA values in male and female mice, indicating that incorporation of the heavy isotope was not sex dependent (SI Appendix, Fig. S2A and B). We observed a strong correlation between protein stability in male and female mice (Fig. 2A). Interestingly, there was an overall trend toward proteins being more stable in male mice. In fact, female mice consumed significantly less food in both the labeling and chase phases (SI Appendix, Fig. S2C and D). A subset of proteins appeared to be differentially stable in male and female mice. We considered proteins differentially regulated by sex as those that had a male/female half-life ratio of greater or less than 2 SD of the mean (Fig. 2A and SI Appendix, Table S2). We noticed that several proteins exhibiting sex differences in their stability were also ASD risk genes, as designated by the SAFARI database (<https://gene.sfari.org/>). STRING analysis of these proteins with sex differences revealed significantly connected proteins, with clusters relating to inhibitory synapses and signal transduction by L1 (Fig. 2B). We considered if X-linked genes were more likely to have differences in their turnover rates in male and female animals. In line with the data from all genes, there was a strong correlation between the turnover rates of X-linked genes (SI Appendix, Fig. S2E).

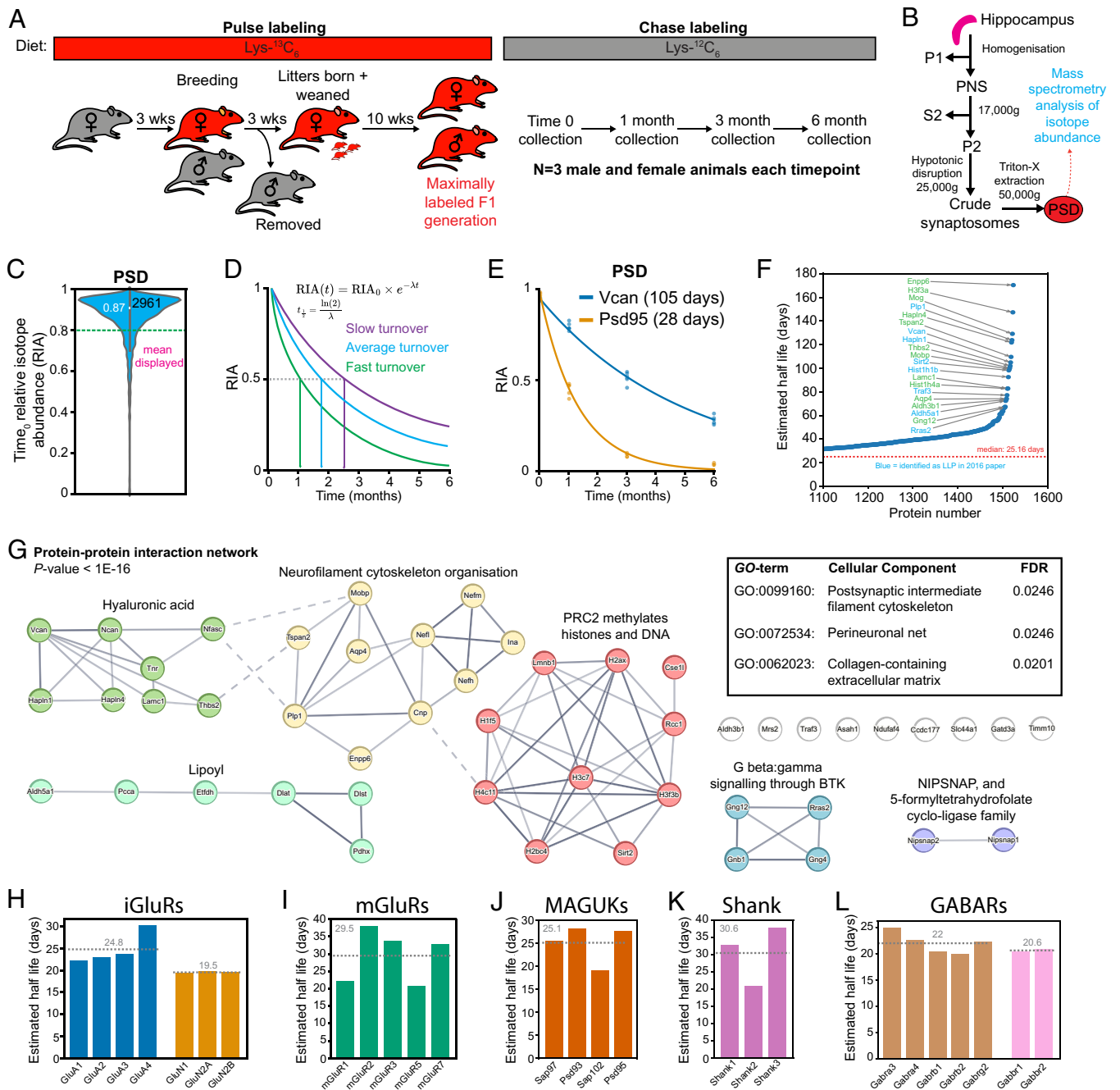


Fig. 1. Identification of stable synaptic proteins. (A) Schematic of pulse chase SILAM labeling scheme. (B) Schematic of tissue processing before mass spectrometry. (C) Violin plot displaying the RIA values for each protein at the beginning of the chase period (Time₀). The green dotted line shows the threshold for inclusion (>0.8) and the white point displays the mean (0.87). (D) Schematic showing fit of exponential decay of the RIA values over time, with examples of curves for slow, average, and fast turnover. (E) Data showing example of proteins with average (Psd95, 28 d) or stable (Vcan, 105 d) estimated half-lives. Each data point represents an individual animal. (F) Scatter plot of individual proteins arranged by relative stability. The top 30 proteins are labeled. Blue labels indicate proteins we previously described as stable synaptic proteins (4). (G) STRING analysis of the top 50 stable proteins. Proteins are colored by their associated cluster labels. A significant protein-protein interaction network indicates that proteins are more connected than would be expected by chance. *Inset* box shows GO-terms for cellular components that were significantly enriched in these 50 proteins compared to a background of all proteins in our PSD dataset. (H-L) manual curation of the estimated half-lives of proteins in key receptor or scaffolding protein families. The gray dotted line indicates the average turnover rate for each group.

Aversive Experience Tunes Stability of Synaptic Proteins. We next sought to assess changes in hippocampal PSD protein stability following cFC. cFC is a well-established hippocampus-dependent task that triggers long term memory following a single training session (Fig. 3A). We confirmed that cFC induces long-term memory in a pilot study with unlabeled animals, as demonstrated by freezing behavior when returned to the conditioning context. In

this cohort, aversive memories persisting up to 2 mo compared to control animals that received no foot shock in the arena (Fig. 3B).

Subsequently, we repeated this protocol with labeled male assigned to control or cFC groups. On the conditioning day, animals assigned to the cFC group began freezing after foot shocks were delivered (SI Appendix, Fig. S3A). Tissue was collected from control or cFC mice at various timepoints: immediately after

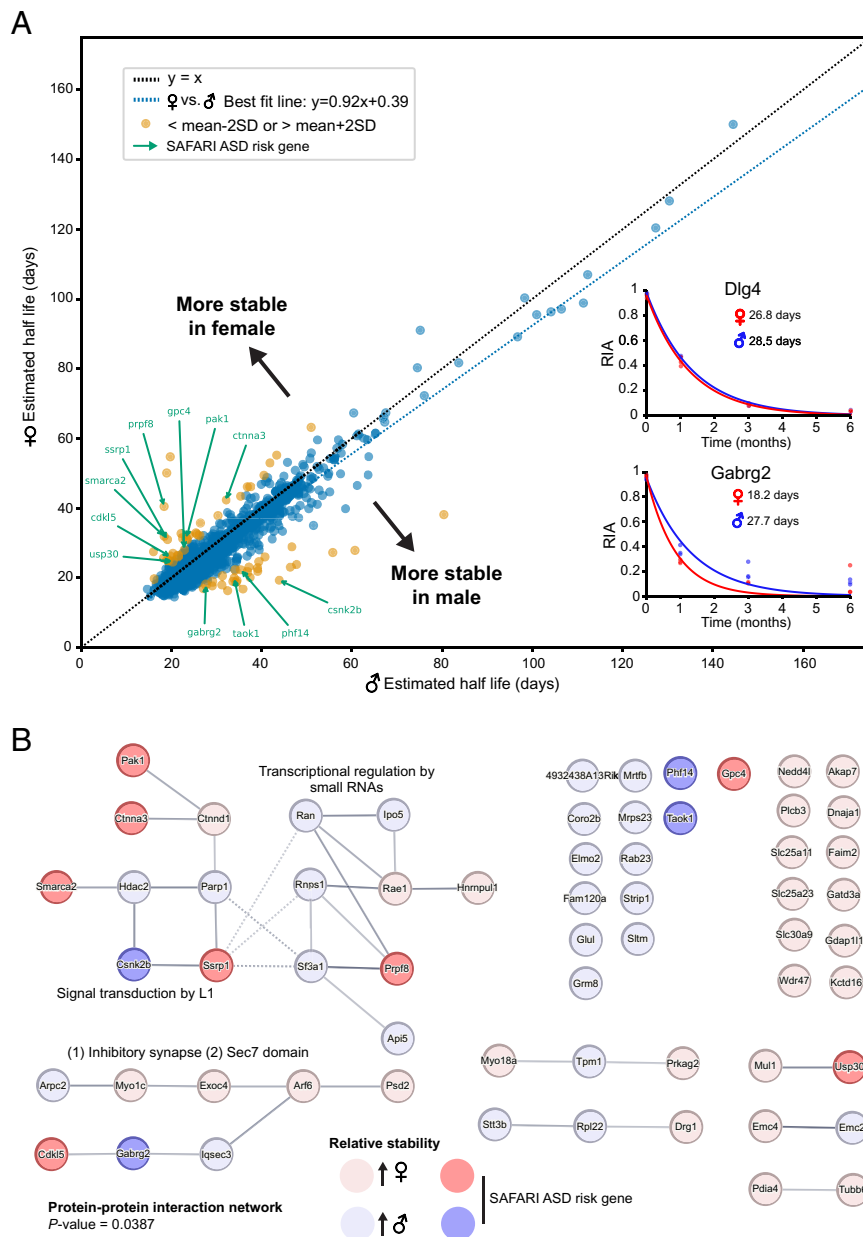


Fig. 2. Sex differences in protein turnover. (A) Scatter plot showing the relative turnover rates of individual proteins in male (x-axis) and female (y-axis) animals. Yellow symbols indicate proteins that were ± 2 SD from the mean. Green arrows and labels identify putative ASD risk genes from the SAFARI database. *Inset* graphs indicate raw data from two example proteins that do not change with sex (Dlg4) or that show greater stability in male mice (Gabrg2). Note, the black dotted line indicates unity between the axes. The blue dotted line shows the best fit of the data, highlighting that proteins were, in general, slightly more stable in male animals. (B) STRING analysis of the proteins with ± 2 SD differences in turnover rates. Blue shading indicates greater stability in male animals and red shading indicates greater stability in female mice. Darker shading in blue or red indicates putative ASD risk genes.

training (to serve as the baseline), or following re-exposure to the arena at 1 wk, 1 mo, or 3 mo. The PSD fraction was then isolated and subject to mass spectrometry to determine the RIA of each protein (Fig. 3 C and D). As described above, we fit an exponential decay to the RIA values measured for each protein over time to estimate the relative half-lives (as in Fig. 1D). The turnover rate of most proteins did not change following FC, showing that there are no global changes in protein turnover. Supporting this, the food intake was not changed in the “chase” period in control and FC animals (SI Appendix, Fig. S3C). However, there were some exceptions that appeared to be differentially regulated following FC (Fig. 3E). We considered proteins differentially regulated if the FC/control half-life ratio was greater or less than 2 SD of the mean (Fig. 3E and SI Appendix, Table S3). STRING analysis of these differentially regulated proteins revealed a significant

protein–protein interaction network, with clusters of proteins related to regulation of dendritic spine morphogenesis and CRMPs in Sema3A signaling (SI Appendix, Fig. S3D).

Subsequently, to evaluate potentially subtle changes in turnover rates of related proteins we examined the FC/control turnover ratios of proteins in our dataset linked to specific Gene Ontology (GO) biological processes, cellular components, and molecular functions (Fig. 3F). We only included terms when our dataset included >20 proteins and conducted one-sample t tests against a FC/control ratio of 1 (i.e., no change). Subtle changes were observed, with groups of proteins related to “learning,” “synaptic organization,” “dendritic spine,” and “PDZ-domain binding” showing elevated stability following FC, and terms such as “positive regulation of protein phosphorylation” showing decreased stability (Fig. 3F). Curiously, there was no correlation between

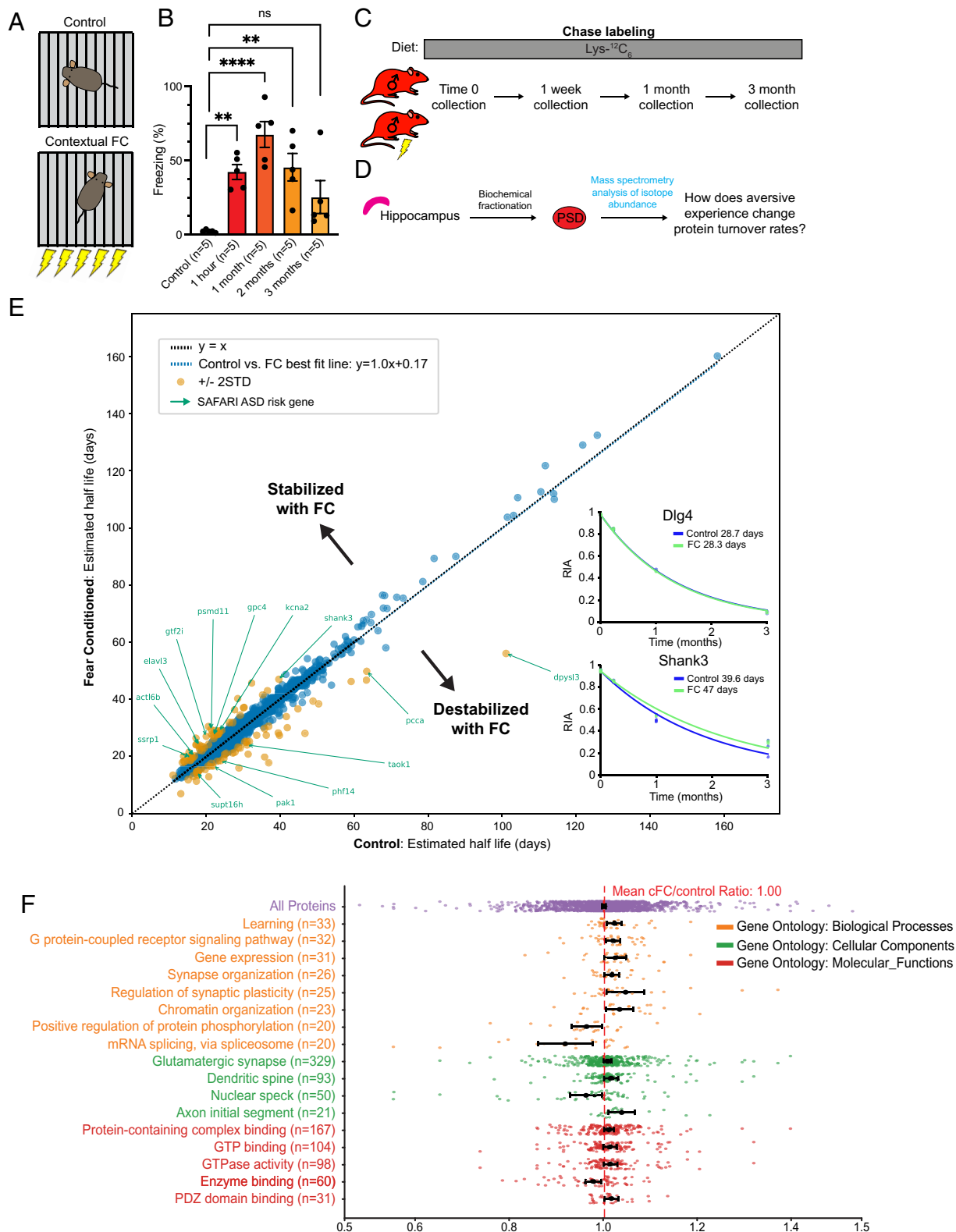


Fig. 3. Aversive experience tunes stability of synaptic proteins. (A) Schematic of the cFC paradigm. (B) Pilot study in unlabeled mice demonstrating robust freezing upon re-exposure to the context, used as a proxy for the fear memory. All groups were compared to the control group, which did not receive foot shocks while in the apparatus. (C) Labeling strategy and time course for tissue collection. Note, in this experiment only male mice were included. (D) Hippocampal tissues were subject to biochemical fractionation and subsequent mass spectrometry to evaluate how experience changes long-term protein stability. (E) Scatter plot showing the relative turnover rates of individual proteins in control (x-axis) and cFC (y-axis) animals. Yellow symbols indicate proteins that were ± 2 SD from the mean. Green arrows and labels identify putative ASD risk genes from the SAFARI database. Inset graphs indicate raw data from two example proteins that do not change with experience (Dlg4) or that show greater stability following cFC (Shank3). Note, the black dotted line indicates unity between the axes. The blue dotted line shows the best fit of the data, highlighting that protein turnover rates were, in general, unchanged with cFC. (F) Plot showing the relative change in turnover rates (cFC/control) for all proteins (purple), or proteins linked to GO biological processes (orange), cellular components (green), or molecular functions (red) that were significantly different to a ratio of 1 (one-sample t test) and have 20 or more protein under the GO terms (for statistical details see *SI Appendix, Table S4*). Error bars display the SEM. The red dotted line indicates no change in turnover rates with cFC.

overall protein stability and the FC/control ratio (*SI Appendix, Fig. S3E*).

Discussion

Unlike other biomolecules that exhibit longevity, such as DNA, proteins do not have the same repair mechanisms. LLPs are therefore predicted to be particularly vulnerable to accumulating oxidative damage over time, posing an inherent risk to cells that maintain LLPs. Consequently, we hypothesized that LLPs must serve an important function to balance out the risk that accumulation of damaged proteins could impart. The goal of this study was to first identify novel LLPs, and then to assess sex and experience dependence changes in protein turnover rate.

Presently, the role of exceptionally stable LLPs remains largely unknown but could explain how certain synapses exhibit remarkable stability in vivo. We used an unbiased approach to evaluate the relative turnover rates of hippocampal proteins enriched in the PSD, a subcellular compartment essential for synaptic transmission and synaptic plasticity. As expected, based on related studies (3, 4), we identified a subset of particularly stable PSD synaptic proteins. Protein turnover rates in the hippocampal PSD (this study) correlated well with previous reports from the synaptosome fraction of the cortex and cerebellum (3), indicating that stability was preserved between brain regions. Several of the LLPs we identified were extracellular matrix proteins such as Versican, Neurocan, and hyaluronan and proteoglycan link proteins 1 & 4. While not components of the PSD, we expect these proteins are in close association with the PSD and remain in complex throughout the PSD fractionation protocol. Within synaptic protein families we observed differences in the turnover rates of individual proteins. For example, of the AMPA-type glutamate receptors, the GluA4 subunit which is particularly enriched in GABAergic inhibitory interneurons (15, 16) exhibited the slowest relative turnover rate. Parvalbumin positive inhibitory interneurons have been shown to have more stable glutamatergic synapses than their counterparts in excitatory neurons (17). It will be fascinating to uncover the molecular basis for the cell type-specific regulation of synapse stability, and to determine if LLPs contribute to such cellular specializations. Within the MAGUK family, Sap102 had the fastest relative turnover rate and is also known to be substantially more mobile at glutamatergic synapses (12). However, we acknowledge that the overall stability of a protein need not match its mobility within a neuron. A PSD protein that could be of particular interest is TNF alpha receptor associated factor 3 (Traf3). A role for Traf3 at the synapse has not been documented, but it has been identified in Psd95 protein complexes previously (18). Furthermore, we recently identified Traf3 as being enriched in Psd95 complexes originating from GABAergic inhibitory interneurons (19). Interestingly, recent GWAS have linked Traf3 to major depressive disorder (20), suicide (21), and posttraumatic stress disorder (22). How the stability of Traf3 could relate to its function at the synapse remains to be investigated.

Sex differences in the prevalence of psychiatric disorders are well documented. For example, ASD is much more readily diagnosed in males (10), and major depressive disorder is more prevalent in females (23). Previous studies evaluating protein turnover rates have not considered sex as a variable. In the present study, we identified a subset of proteins that were differentially stable in male and female mice. Curiously, a number of these proteins were putative ASD risk genes, such as Cdkl5 and Taok1. Cdkl5 deficiency disorder (CDD) is rare and associated with early-onset seizures and severe intellectual disability (24). CDD cases are more predominant in females (25), possibly as the gene is X-linked.

Cdkl5 is enriched at the PSD in mature neurons and knockdown was shown to reduce the number of functional synapses (26). Spontaneous seizures have been reported in aged female mice with Cdkl5 loss of function (27), but not in younger male mice (28, 29). It would be interesting to test if younger female mice also exhibit spontaneous seizures, as we observed that Cdkl5 was more stable at the synapse in female mice. We found that the ASD risk gene, Taok1, was more stable in the PSD of male mice. Interestingly, haploinsufficiency of Taok1 in a mouse model leads to behavioral changes relevant to ASD (30), with deficits in social approach only observed in male animals (30). In the present study, Shank3 was on the borderline of being considered a protein more stable in male mice (*SI Appendix, Table S2*). This is potentially interesting as Shank3 KO male mice exhibit a more pronounced decrease in hippocampus volume compared to female animals (31). In addition, while the phenotypes in many behavioral assays were similar, on some motor tasks male Shank3 KO mice were more susceptible compared to female animals (32). Furthermore, in a Shank3 model with a c-terminal mutation, it was male mice that exhibited the greatest number of abnormalities in tests of gait, accompanied by more substantial loss of Purkinje cells, and mGluR5 protein (33). Collectively, these examples highlight how loss, or mutation to, proteins more stable in one sex appear to impact that sex on assays of brain function to a greater effect. This potentially indicates a greater reliance on these proteins for proper function, which, we speculate, is why they are more stabilized at the synapse. Considering this, we posit that sex differences in protein stability—if also present in humans—could contribute to sex-differences in the prevalence of psychiatric disorders. More work is required in this area, but our data clearly highlight the necessity to consider sex-differences when evaluating protein turnover rates.

A major goal of our study was to identify proteins whose turnover rates change following learning. We picked cFC as a learning paradigm at this task produces robust one-trial learning. Overall, the average turnover rate of PSD proteins did not change with cFC, but there were subsets of proteins that were stabilized after cFC. Interestingly, the overall stability of a protein did not correlate with its likelihood of changing stability following cFC, showing that these changes cannot be predicted with information about the baseline turnover rate, and instead need to be empirically calculated. It remains unclear what properties of a protein predicts its likelihood to change stability with experience.

One example of a stabilized protein following cFC is Shank3, a protein that is well documented to play a role in synaptic plasticity and learning and memory (34). Therefore, stabilizing Shank3 following cFC could serve as a potential mechanism to promote long-term synaptic potentiation, supporting learning. However, we would like to highlight a caveat to this interpretation as we are currently unable to disambiguate between “learning-dependent” vs. “experience-dependent” changes in protein turnover. For example, it is possible that some of the changes we observed are a result of the aversive experience and stress induced by the foot shocks rather than learning per se. Future studies could include a shock only group to help to account for this possibility.

With respect to the molecular basis of long-term memory, particular attention has been given to extracellular matrix proteins that form perineuronal nets around neurons and synapses as it has been postulated that changes in perineuronal nets could serve as a stable scaffold supporting long term memory (35). While composed of very stable proteins, the extracellular matrix retains the ability for highly dynamic changes, creating time windows permissive for synaptic plasticity (36). In the present study we did not observe experience-dependent changes in ECM protein turnover rates that might have been expected based on the role of ECM

remodeling in learning and memory. However, it is possible that evaluating the entire hippocampus masks subtle changes that could be restricted to specific inputs or synapse populations. Recently a new approach has been utilized to track the stability of individual proteins over time using HaloTag-labeled endogenous proteins in conjunction with photostable dyes. This enables time locked labeling of proteins with distinct fluorescent dyes to map the turnover rates while also preserving spatial information. For example, this has been done with GluA2 and Psd-95 (37, 38). This has shown that in older mice, Psd-95 turnover is slower (37), and that the ionotropic glutamate receptor GluA2 shows region specific increases in turnover following acquisition of a new rule in a spatial learning task (38). In the future, using halo-tagged KI mice for specific ECM proteins—and other candidate proteins—in pulse/chase experiments could provide a sensitive way to map how protein turnover could be regulated in a manner to support synaptic plasticity.

Limitations of the Study

Our approach was tailored toward detecting proteins with slow rather than fast turnover rates. A modified experiment design with closer spaced timepoints would be useful in evaluating proteins with faster turnover rates, which could also play important roles in regulating synapse function. In addition, our approach lacks the spatial resolution for where changes in protein turnover rates are occurring. As a result, we recognize that if turnover rates change in opposite directions in different subregions of the hippocampus, then these changes could be averaged out and missed in our analysis. This information could be determined with generation of HaloTag-KI proteins and sequential labeling with fluorescent dyes. Finally, in our experiment to evaluate experience-dependent changes in protein turnover we only evaluated changes in male mice. In future studies it will be important to evaluate if similar proteins are stabilized or destabilized in female animals, especially as we reveal sex-dependent changes in protein turnover at baseline in this study.

Materials and Methods

Animals. C57Bl/6J animals were purchased from (Jackson Laboratory) and either used to establish a behavioral paradigm that induces long-term learning and to set up breeding pairs for the SILAM experiment. Animal care, use, and experimental protocols were approved by the Institutional Animal and Use Committee (IACUC) of Johns Hopkins University.

SILAM. Female C57Bl/6J mice were switched to a “heavy diet” (Cambridge Isotope Laboratories, Lys-¹³C₆) for 3 wk before being paired up with male C57Bl/6J animals. Male animals were removed when females were visibly pregnant, or at 3 wk. When litters were born, and after weaning, mice remained on the “heavy diet” until 10 wk old for the pulse labeling phase, at which point tissue was collected from a subset of the mice, and food switched to a “light diet” (Cambridge Isotope Laboratories, Lys-¹²C₆) for the chase labeling phase. When 10 wk the animals were allocated into cohorts for two experiments, which were run at the same time:

Experiment 1: Sex specific changes in protein stability. Tissue from male and female animals (n = 3–4 per sex/time point) was collected at 0, 1, 3, and 6 mo following the switch to “light diet.” Data from these groups were used to identify LLP and to evaluate sex differences in protein turnover. Note: all animals in experiment 1 experienced a brief exposure to the fear conditioning chamber at time point 0, with no shock delivery so they could serve as controls for *Experiment 2* to maximize use of valuable tissue.

Experiment 2: Experience-dependent changes in protein stability. To evaluate how aversive experience can influence protein turnover rates, an additional group of male animals underwent contextual fear conditioning (FC; details below) immediately before being switched to the “light diet” to commence the chase period. The hippocampus was then dissected from these mice at week 0 (baseline, no switch

to “light diet”), 1 wk, 1 mo, and 3 mo post FC during the chase period. As a control, mice that were exposed to the fear conditioning chamber—but did not receive any shocks—were used with tissue collected at the same week 0, or 1 wk, 1 mo, and 3 mo timepoints. The control group’s exposure to the apparatus without shock was important to account for biological changes triggered by exposure to a novel environment alone, and to allow us to evaluate changes linked to fear conditioning. In addition, in contrast to *Experiment 1*, we included a 1-wk time point to help capture early changes in protein stability following the aversive experience of fear conditioning that might be missed using only the 0, 1, and 3 mo time points alone.

cFC. Ahead of testing mice were transitioned to a reverse light cycle room and all behavioral testing was conducted during the dark (active phase). Mice were habituated with gentle handling for the days preceding testing. Animals were placed into fear conditioning chambers for 300 s and given 120 s to acclimatize to the arena. Mice in the fear conditioning group (FC) received 5 foot shocks (0.75 mA, 2 s) delivered at 120 s, 150 s, 180 s, 210 s, and 240 s and were returned to their home cage. Control mice (Con) were placed in the same arena for 300 s but no shocks were delivered. This provided a control for context exposure. At different timepoints, animals were re-exposed to the same arena and their freezing was measured over the course of a 300 s recall trial. Freezing was considered as a measure for learning.

Establishing the task: Male C57Bl/6J mice aged 10 wk were allocated into different groups: Con, 1 h FC, 1 mo FC, 2 mo FC, and 3 mo FC. 5 mice were present in each group. Control group data were from the recall session at 1 h.

During the SILAM experiment: SILAM labeled male mice were assigned into Con or FC groups. Tissue was collected at time point 0 (4 mice, 2 Con, 2 FC) or immediately following the recall session after 1 wk (6 mice, 3 Con, 3 FC), 1 mo (6 mice, 3 Con, 3 FC), 3 mo (6 mice, 3 Con, 3 FC), and 6 mo (6 mice, 3 Con, 3 FC).

Biochemical Fractionation. At the respective timepoints the hippocampus was rapidly dissected, and flash frozen in liquid nitrogen. Samples were stored at –80 °C. As previously described (39), the hippocampus from each mouse was homogenized individually using 20 strokes from syringes equipped with 26G × 3/8 (0.45 mm × 10 mm) needles in homogenization buffer [320 mM sucrose, 5 mM sodium pyrophosphate, 1 mM EDTA, 10 mM HEPES pH 7.4, 200 nM okadaic acid, 1 mM sodium orthovanadate, protease inhibitor cocktail (Roche), phosphatase inhibitor cocktail (Sigma-Aldrich)]. The homogenate was then centrifuged at 1,000 × g for 10 min at 4 °C to yield P1 (nuclear fraction) and postnuclear supernatant (PNS) fractions. PNS fraction was further centrifuged at 17,000 × g for 20 min at 4 °C to yield P2 (membrane/crude synaptosome) and S2 (cytosol) fractions. P2 was resuspended in hypotonic resuspension buffer [Milli-Q® water with 5 mM sodium pyrophosphate, 1 mM EDTA, 10 mM HEPES pH 7.4, 200 nM okadaic acid, 1 mM sodium orthovanadate, protease inhibitor cocktail (Roche), phosphatase inhibitor cocktail (Roche)], then centrifuged at 25,000 × g for 20 min at 4 °C to yield lysed synaptosome (LS) fractions. Collected LS fractions were resuspended in resuspension buffer [50 mM HEPES pH 7.4, 5 mM sodium pyrophosphate, 1 mM EDTA, 200 nM okadaic acid, 1 mM sodium orthovanadate, protease inhibitor cocktail (Roche), phosphatase inhibitor cocktail (Roche)] and then mixed with an equal part of 1% Triton X-100 (containing protease and phosphatase inhibitors). This mixture was incubated at 4 °C with rotation for 10 min followed by centrifugation at 50,000 × g for 20 min at 4 °C to yield PSD preparation. PSD pellets were frozen and subsequently processed for mass spectrometry (see below).

Mass Spectrometry-Based Proteomics. The PSD pellet was reconstituted in 100 µL of lysis buffer (4% SDS in 0.1 M Tris-HCl pH 7.6) and sonicated using a tip sonicator (Branson, SFX 550). Protein concentration was measured using BCA protein assays (Thermo, 23227). Proteins were reduced using 100 mM dithiothreitol for 30 min followed by alkylation with 50 mM iodoacetamide for 30 min. Around 50 µg of proteins were transferred to a 10 kDa centrifugal filter (Millipore Sigma, Amicon Ultra-0.5), in which the protein sample was mixed with 8 M urea (in 0.1 M triethylammonium bicarbonate, TEAB). The proteins on the membrane filter were centrifuged at 4,000 × g for 10 min to remove SDS followed by buffer exchange with 100 mM TEAB. Lys-C (FUJIFILM Wako, 125-05061) was added to the filter at an enzyme to protein ratio of 1: 50 and digested at 37 °C for overnight. Peptides were analyzed using the Q Exactive HF mass spectrometer (Thermo Scientific, San Jose, CA) coupled to Ultimate 3000

liquid chromatography system (Thermo Scientific, San Jose, CA). Around 1 μg of peptides of individual samples were loaded onto a trap column (2 cm $\times$ 100 μm , Acclaim PepMap100, Thermo Fisher Scientific) and separated using 200 cm long micropillar array column (PharmaFluidics). The following gradient was generated with solvent A (0.1% formic acid in water) and solvent B (0.1% formic acid in acetonitrile): 5 to 7% sol B in 15 min, 7 to 27% sol B in 235 min, 27 to 40% sol B in 15 min, 40 to 80% sol B in 20 min, maintaining at 80% sol B for 15 min, and equilibrating with 5% sol B for 20 min. The flow rate was set as 300 nL min⁻¹. Mass spectrometry data were acquired in data-dependent acquisition mode. Full MS scans were acquired for the mass range of 350 to 1,800 m/z at the resolution of 120,000. Twenty most abundant ions were isolated under 1.6 m/z isolation width and fragmented with normalized collision energy (NCE) of 27. The MS/MS scans were acquired at a resolution of 30,000 with a fixed first m/z of 110 m/z. Maximum ion injection time was 50 ms and 54 ms for full MS and MS/MS scan, respectively. The raw mass spectrometry data were searched using Andromeda in MaxQuant software suite (version 2.0.1.0) against a UniProt mouse protein database (17,015 entries) along with common contaminants. Peptides with a maximum of two missed cleavages were considered. Acetylation of protein N-terminus, oxidation of methionine, and deamidation of asparagine and glutamine were set as variable modifications and carbamidomethylation of cysteine was used as a fixed modification. False discovery rate was applied as 1%

Further analyses were conducted to compare protein half-lives both between sexes and between behaviorally manipulated experimental groups. For the sex-specific comparison, the primary dataset described in previous step was stratified into male and female cohorts. These cohorts were then analyzed separately, applying the identical analytical model, fitting strategies, and inclusion criteria used for the primary analysis. Samples for these sex-specific analyses were collected at 0 wk ($n = 4$ for males, $n = 3$ for females), and at 1 mo, 3 mo, and 6 mo ($n = 3$ for both males and females at each of these subsequent timepoints).

For comparisons involving behavioral manipulation, fear-conditioned (FC) male mice were compared to control male mice. Samples for this analysis were collected at 0 wk ($n = 4$ for both FC and control groups), and at 1 wk, 1 mo, and 3 mo ($n = 3$ for both groups at these subsequent timepoints). Crucially, for all these group-specific comparisons, protein half-lives were estimated employing the identical analytical formula (the single exponential decay model described previously), nonlinear least squares regression fitting strategy, and all associated data and fit quality inclusion criteria as detailed for the primary analysis.

For select validation experiments, a modified double exponential decay function incorporating heavy amino acid recycling constants was applied to refine lifetime estimations (3). This model accounts for two distinct recycled pools and was adapted from published formulations describing amino acid turnover in vivo:

$$\text{RIA}(t) = 1 - e^{-\frac{t}{t_{1/2}}} \times \left(1 - \frac{0.503 \times \frac{0.799}{\ln(2)} \times \left(1 - e^{-\frac{t}{t_{1/2}} - \frac{0.799}{\ln(2)}} \right)}{\frac{0.799}{\ln(2)} - t_{1/2}} - \frac{(1 - 0.503) \times \frac{39.423}{\ln(2)} \times \left(1 - e^{-\frac{t}{t_{1/2}} - \frac{39.423}{\ln(2)}} \right)}{\frac{39.423}{\ln(2)} - t_{1/2}} \right)$$

for both peptide and protein level. Quantification of SILAC pairs was performed with standard settings in MaxQuant.

The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE (40) partner repository with the dataset identifier PXD065808.

Mass Spectrometry Data Analysis. The matrix of relative intensities of identified peptides from MaxQuant was processed using Python (SciPy v1.13.1, NumPy v1.26.4). For each peptide, the relative intensities corresponding to light (Lys-¹²C₆, L) and heavy (Lys-¹³C₆, H) lysine isotopes were extracted, and the RIA was calculated for each individual protein using the formula:

$$\text{RIA} = \frac{H}{H + L}$$

RIA values were determined for each protein across four experimental timepoints (0 d, 1 mo, 3 mo, and 6 mo), with biological replicates consisting of 4 males and 3 females at day 0 ($n = 7$) and 3 males and 3 females at each subsequent timepoint ($n = 6$). For each protein, the RIA at day 0, denoted as RIA_0 was calculated from these initial measurements, and protein turnover rate ($t_{1/2}$) were subsequently estimated by fitting the RIA values for each protein to a single exponential decay model. This model employed the precalculated RIA_0 as a fixed initial abundance instead of 100%, with the best-fit protein turnover rate determined via nonlinear least squares regression (implemented using SciPy's "curve_fit" function), according to the following equation:

$$\text{RIA}(t) = \text{RIA}_0 \times e^{-\frac{\ln(2)}{t_{1/2}} \times t}$$

Following model fitting, proteins and their corresponding estimated half-lives were selected for downstream analysis only if they satisfied a set of stringent data and fit quality criteria: a coefficient of determination between fitted curve and the observed data of at least 0.8 ($R^2 \geq 0.8$), average day 0 RIA greater than 80% ($\text{RIA}_0 \geq 0.8$), no more than three missing values across all timepoints, and at least one valid RIA measurement at each timepoint.

Pathway Analysis. All proteins passing filtering criteria in the combined dataset ($n = 1,526$) were annotated via automated API calls to the Universal Protein Resource (UniProt Consortium) database hosted by EMBL-EBI (41). For each protein, associated Ensembl gene identifiers, sequence data, functional domains, and GO annotations were retrieved. The Uniprot and Ensembl identifiers were used to cross-reference additional datasets, including the SFARI Gene database for ASD risk genes (42). Human ASD risk genes were retrieved, and their associated risk scores were matched with our half-life dataset. Genes were considered ASD risk genes if they had an ASD risk score of 1–3 or were classified as syndromic genes.

To assess functional enrichment and protein–protein interactions, STRING network analysis was performed with custom background genes of all the proteins identified in mass spectrometry (43). The top 50 most stable proteins were subjected to unsupervised clustering using k-means ($k = 6$) to identify cluster labels, and significantly enriched nonredundant GO terms are extracted. STRING analyses were also carried out on proteins exhibiting sex-specific differences in stability (defined by female-to-male half-life ratios exceeding ± 2 SD from the overall mean) and proteins with tuned stability after aversive experience (defined by cFC/control half-life ratios exceeding ± 2 SD from the mean), each clustered using k-means ($k = 8$). All STRING analyses significance of interaction networks was determined based on default STRING enrichment thresholds.

Data, Materials, and Software Availability. Proteomics data have been deposited in ProteomeXchange consortium (identifier PXD065808) (40).

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